

Akshay Putcha

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Education

Texas A&M University | College Station, TX
Computer Engineering | 05/2029
GPA: 3.3/4.0

Projects

AI-Powered Product Ingredient Scraper

- Designed and built a cross-platform desktop app (Tauri + React/TypeScript frontend, Python/FastAPI backend) that automates ingredient-list extraction from cosmetics/chemical product pages, replacing analysts' manual copy-paste workflow.
- Architected a 5-tier extraction cascade (structured data parsing → CSS heuristics → LLM extraction → agentic browser automation → AI web search) that escalates cost only on failure, minimizing per-product API spend.
- Integrated the Anthropic Claude API (Sonnet agent with constrained tool-use, Haiku for cheap extraction) with Playwright browser automation, including prompt caching and hard cost/turn budgets to control per-job spend.
- Implemented a human-in-the-loop review UI (approve/edit/reject extracted data, live SSE job progress, cost tracking) and Excel export/analytics dashboard, backed by a local SQLite store.

Document Search Engine

- Built LocalLens, a local-first document search engine in **C++20/Qt 6** that indexes and full-text searches folders offline, using **BM25 ranking** for relevance and returning results in microseconds
- Designed a multi-threaded architecture (UI, filesystem watcher, and indexer worker threads) using **Qt's queued signal/slot system** and a **mutex/condition-variable task queue** to keep the UI responsive under continuous background indexing
- Implemented a native filesystem watcher using Win32's ReadDirectoryChangesW for real-time re-indexing without polling, plus a portable QFileSystemWatcher fallback

Environmental Monitor

- Built Envi-M, a full-stack environmental monitoring system combining **Arduino (C/C++)** firmware with a **Python** data pipeline to score how conducive a space is for focused work
- Wrote embedded firmware to read **temperature (DHT11)**, **humidity**, and **light intensity (photoresistor)** sensors and stream readings over Serial to a host machine
- Built a Python script to parse the serial stream, log readings to **CSV** with timestamps, and compute a custom "Focus Score" from threshold-based penalties (temp, humidity, light)

Experience

One Sight Technology | Tamil Nadu, Chennai
Junior Software Developer | 05/2024 - 05/2025

- Spearheaded the technical design mobile application prototype aimed at improving accessibility and quality of life for children with Autism Spectrum Disorder (ASD).
- Engineered customized UI/UX features, including specialized color modes, to maximize user engagement and accessibility.
- Architected and implemented a backend storage solution to track user preferences and save progress across gamified learning modules.

Skills

- Frameworks & Tools:** Python, C++, Java, Claude Code in Action, Building with Claude API, Java Certificate.